Double-Phase Hologram Steganography using Keyed Gabor Splatting Representations: supplemental document

This supplemental document provides an extended review of related work, exact evaluation definitions and implementation settings, Quantization Index Modulation (QIM) calibration and payload statistics, the carrier-selection and recovery validation, objective details, and cross-dataset validation results.

S1. RELATED WORK

S1.1. Double-Phase Hologram Generation and Explicit Representation

Phase-only computer-generated holograms encode a target complex-valued optical field into a phase pattern that forms the desired image after propagation. Double-phase encoding represents such a field using two phase-only components. In the spatial Double-Phase Hologram (DPH) construction used in this work, the two components are interleaved on a checkerboard to form a single phase-only hologram [1]. This alternating structure introduces strong local phase variation and makes DPHs challenging to represent using only smooth image primitives [2].

Multi-color hologram generation has been studied to improve holographic-display brightness and image quality through multiple color primaries [3]. Coordinate-based and autoencoder models have also been evaluated for phase-only hologram compression [4]. Most directly related to our representation pipeline, Fan et al. [2] unpacked the two checkerboard branches of a DPH and fitted them using an additive GaussianImage-style representation. GaussianImage represents two-dimensional signals using accumulated anisotropic Gaussian primitives [5], whereas Gabor Splatting (GS) augments localized Gaussian envelopes with periodic modulation to better represent localized high-frequency image structure [6]. Collectively, these studies address hologram generation and compression or provide explicit signal representations. We instead study whether a representation fitted to a clean DPH can serve as a post-generation payload carrier while preserving the propagated optical reconstruction.

S1.2. Holographic Information Hiding

Digital holography has been used to encode hidden images before embedding them into conventional host images. Hamam [7] constructs a phase-only or complex hologram of a secret image and stores its quantized values in the low-order bits of an ordinary cover image. Zou et al. [8] further assign pixelwise embedding capacity using the host intensity and saliency map, preserving salient regions while embedding a multilevel phase-only hologram. In both approaches, the hologram represents the hidden image, whereas the actual steganographic carrier is a conventional intensity image.

Digital holograms have also served directly as watermark carriers. Prior methods optimize watermark strength in hologram data, embed authentication signals in transformed hologram coefficients, or learn hologram-domain embedding and extraction while accounting for reconstructed image quality [9–11]. Shimobaba et al. [12] convert reduced hidden information to a complex amplitude, add it to the host field, and then generate a hologram in which the embedded reconstruction is visually inconspicuous. Other work incorporates auxiliary data during hologram generation [13].

Other studies make holograms part of authentication, encryption, or physical storage systems. Wang et al. [14] jointly generate two camouflaged computer-generated holograms (CGHs) whose individual diffraction patterns are combined to derive the sampling path for Fourier ghost imaging (FGI) decryption, while their cascaded diffraction pattern provides holographic authentication. Jiang et al. [15] use a dual-cation optoelectronic memory to electrically hide and reproduce holographic patterns, with unit locations and voltage-pulse sequences serving as independent spatial and temporal retrieval keys. These methods operate on conventional image hosts, hologram-domain values or transform coefficients, generation-time encodings, or physical

Table S1. Method-specific serialization rules used in the reported analyses.

Method	Canonicalization	Quantization
GS	**Periodic wrapping**	**Floor**
GaussianImage	Clipping	Nearest integer
INRSteg, RoSteALS	Clipping	Floor

holographic media. We instead begin with an already generated display-oriented DPH, fit it with an explicit Gabor representation, and embed a bit payload after generation by modifying saved primitive parameters. We evaluate the modified DPH through holographic reconstruction and recover the payload from the saved representation.

S1.3. Learned and Parameter-Domain Information Hiding

QIM assigns host features to message-dependent quantization cosets [16]. White-box neural-network watermarking establishes model parameters as writable carriers for ownership information, although its task and model class differ from ours [17]. Most closely related, Wu et al. [18] embed keyed payloads in saved 2D Gaussian image parameters and recover them from the parameter set. Foveated steganography increases the capacity of latent image representations using a perceptually guided objective [19]. Separately, recent 3D Gaussian Splatting watermarking methods modify or densify explicit scene representations and recover ownership messages from Gaussian parameters, rendered views, or both [20–22]. Our setting instead uses two-dimensional GaussianImage and GS representations of a DPH and evaluates their effect after optical propagation. INRSteg uses implicit neural representations for cross-modal hiding [23], and we adapt its coordinate multilayer perceptron (MLP) formulation to the DPH benchmark. RoSteALS embeds a message through the latent space of a pretrained image autoencoder and decodes it from the resulting stego image [24]. In contrast, the evaluated GaussianImage and GS implementations recover bits from the saved representation and the shared key. Because these methods decode from different artifacts, we report their bit-error rates within each decoder domain rather than interpret them as equivalent robustness measurements.

S2. EVALUATION ENDPOINTS AND METRICS

S2.1. Phase serialization

For a physical phase prediction φ in radians, let $x = \varphi/(2\pi) \in \mathbb{R}$ denote its cycle-normalized form, where one unit is one complete 2π cycle. The experiments use periodic or clipped canonicalization and floor or nearest-integer byte quantization:

$$c_{\mathrm{per}}(x) = x - \lfloor x \rfloor, \qquad c_{\mathrm{clip}}(x) = \min(1, \max(0, x)), \tag{S1}$$

$$Q_{\mathrm{floor}}(c) = \lfloor 255c \rfloor, \qquad Q_{\mathrm{round}}(c) = \min\big(255, \lfloor 255c + \tfrac{1}{2} \rfloor\big)\,. \tag{S2}$$

The implementation uses cycle-normalized phase x for fitting and serialization. Its circular objective $1 - \cos[2\pi(F - x)]$ is equivalent to the radian-domain formulation in the main text. Table S1 gives the method-specific mappings used throughout the reported analyses. All stored-phase and optical evaluations use the resulting 8-bit phase images.

A stored byte q is decoded as

$$\varphi_q = 2\pi q/255, \qquad U_q = \exp(i\varphi_q), \tag{S3}$$

before holographic reconstruction. Stored-phase metrics compare the byte-domain phase maps after the method-specific mappings above. Focused-plane metrics are computed after decoding, inserting the candidate patch into the clean full hologram, and propagating the complete hologram.

S2.2. Phase and optical metrics

Cross-method phase comparisons and qualitative annotations use the aligned metrics below. Figure S1 retains serialized phase peak signal-to-noise ratio (PSNR) only as a within-method embedding-distortion diagnostic.

Following Choi et al. [25], Phase Distribution Alignment (PDA) aligns phase distributions before PSNR and structural similarity index (SSIM) evaluation. The resulting scores are PDA-PSNR and PDA-SSIM. We use the authors' public implementation, pinned to commit `f0a817b`. We pass each recomposed 512×512 DPH phase-patch channel to the unmodified functions as float64 values in the 0–255 scale. For continuous fits, we multiply the canonical cycle-normalized phase by 255 without quantizing. For stored outputs, we use the original serialized bytes. The code performs double mean removal with modulo-256 wrapping, a conditional 128-level shift, and relative mean-offset alignment before conversion to phasors. This metric-domain modulo-256 convention is distinct from the $q/255$ optical decoding in Sec. S2.1. PDA-PSNR sums cosine and sine squared errors and uses peak value $2\sqrt{2}$. PDA-SSIM combines the two coordinate scores by their root mean square, using the implementation defaults: a 7×7 uniform window, sample covariance, and data range 2. Across red-green-blue (RGB) channels and repeated keys, PDA-PSNR pools phasor mean squared error (MSE) before conversion to decibels, whereas PDA-SSIM averages the channel and key scores. We do not align the high- and low-value branches independently. Alignment changes only the metric inputs, not the saved states or patches used for optical evaluation.

For optical evaluation, the stored candidate patch is inserted at its fixed coordinates in the clean full hologram. The clean and candidate holograms are propagated to the intended depth, and metrics use the corresponding 512×512 RGB reconstruction region of interest (ROI). Propagated intensities are clamped to $[0, 1]$. Clean and candidate reconstructions are not independently normalized or aligned. Optical PSNR aggregates pixels and channels in MSE space before conversion to decibels. SSIM [26] uses an 11×11 Gaussian window (standard deviation 1.5), stride one, no padding, and the standard constants $C_1 = 0.01^2$ and $C_2 = 0.03^2$. Learned Perceptual Image Patch Similarity (LPIPS)-Alex [27] uses the AlexNet variant on the same RGB ROI after conversion from $[0, 1]$ to $[-1, 1]$, with no additional normalization inside the metric implementation.

S2.3. Qualitative visualization protocol

The phase-error maps in Fig. S2 show the per-pixel error underlying PDA-PSNR after phase-distribution alignment and use a shared square-root color scale capped at the pooled 99.5th percentile. RGB reconstruction error is the per-pixel mean absolute channel difference and uses a linear scale capped at the pooled 99.9th percentile. We visualize DIV2K 0173 for the representation and objective comparison and CLIC IMG_1115 for the main-paper comparison.

S2.4. Aggregation and uncertainty

The source is the independent statistical unit. Fitting initializations and QIM keys are within-source repeats. Table S2 records the analysis-specific aggregation and inference. Reported means and sample standard deviations are computed across sources. The sensitivity tables retain medians across sources and subset seeds.

Seeds are 3342026 for the four-method PDA contrasts, 3352026 for the clean- representation PDA contrasts, and 20260823 for the dataset-stratified optical comparison. Remaining seeds are recorded in the accompanying reproducibility files. Intervals are pointwise and unadjusted unless stated otherwise. An interval can therefore exclude zero while a Holm-adjusted p-value exceeds 0.05.

All reported evaluations use one deterministic centered patch per source. This removes the content-dependent patch selector from the evaluation protocol. Consequently, the results characterize the fixed central-patch setting and do not establish invariance to arbitrary embedding locations.

S3. IMPLEMENTATIONS AND OPTIMIZATION

S3.1. GaussianImage baseline

For checkerboard component $s \in \{\mathrm{hi}, \mathrm{lo}\}$, GaussianImage uses N_{G} additive anisotropic Gaussian primitives. Primitive i has center $\mathbf{u}_{i,s}$, covariance $\mathbf{\Sigma}_{i,s} = \mathbf{L}_{i,s}\mathbf{L}_{i,s}^{\mathsf{T}}$, and weighted coefficient $\widetilde{\mathbf{c}}_{i,s} \in \mathbb{R}^K$. Opacity is absorbed into the coefficient. The accumulated field is

$$\mathbf{F}_{\mathrm{G},s}(\mathbf{x}) = \sum_{i=1}^{N_{\mathrm{G}}} \widetilde{\mathbf{c}}_{i,s} \exp\left[-\frac{1}{2}(\mathbf{x} - \mathbf{u}_{i,s})^{\mathsf{T}} \mathbf{\Sigma}_{i,s}^{-1} (\mathbf{x} - \mathbf{u}_{i,s})\right]. \tag{S4}$$

Following the DPH-specific rendering rule of Fan et al. [2], the rendered phase component is

Table S2. Aggregation and inference used in the reported analyses.

Analysis	Independent unit	Repeats	Aggregation	Interval or test	Multiplicity
Objective comparison	Source	Matched objectives	RGB phasor MSE for PDA-PSNR and arithmetic mean for PDA-SSIM	95% paired Student-t interval	None
100-bit four-method comparison	Source	Three keys per GaussianImage and GS source, and one INRSteg and RoSteALS output per source (1,600 outputs)	MSE before PSNR, arithmetic mean for SSIM and LPIPS, and PDA as in Sec. S2.2	95% percentile source bootstrap (20,000 draws)	None
Clean representation	Source	One fitted state per method	MSE before PSNR and PDA as in Sec. S2.2	95% percentile source bootstrap (20,000 draws)	None
Payload-size studies	Source	Three keys per source and payload	Keys aggregated in MSE space before PSNR	95% source bootstrap and two-sided paired sign-flip test (20,000 draws each)	Holm across six payloads

$$\widehat{\varphi}_{\mathrm{G},s}(\mathbf{x}) = 2\pi \left[\frac{1}{2} \sin\big(\mathbf{F}_{\mathrm{G},s}(\mathbf{x})\big) + \frac{1}{2} \right]. \tag{S5}$$

We fit GaussianImage using pixelwise phase MSE without optical-propagation supervision. Section S2 gives the equivalent cycle-normalized implementation. By contrast, GS applies no output sine activation. It converts the accumulated field directly to physical phase and wraps it modulo 2π, as defined in the main paper.

S3.2. Factorized versus independent per-atom coefficients

We retain the original GS parameterization [6]: a color vector $\mathbf{c}_i$ and scalar frequency coefficients a_{ik}, with $\mathbf{b}_{ik} = a_{ik}\mathbf{c}_i$ as derived notation (component index omitted). For comparison, we independently optimized one RGB vector $\mathbf{b}_{ik}$ per frequency atom. This removes the shared color direction across a primitive's atoms and increases the trainable parameter count from 768,000 to 1,696,000 scalar values across both components of one 512×512 patch. It is not an equivalent reparameterization.

We fit both forms on five fixed DIV2K sources using one centered patch per source, 16,000 primitives per component, circular supervision, matched initial rendered fields, and the same optimizer schedule and seeds (42/43 for high-/low-value components). Table S3 reports the checkpoint comparison. The factorized form attains higher mean continuous-phase PDA-PSNR at the earlier checkpoints, while the final serialized PDA-PSNR is nearly equal: the factorized form is 0.068 dB higher on average and wins on four of five sources. These results support retaining the factorized parameterization for this pipeline.

This five-source check is descriptive rather than population-level. The two forms were neither capacity-matched nor tuned independently, so the runtime difference cannot be attributed solely to factorization.

S3.3. Implementations and optimization settings

Table S4 lists the pinned sources, adaptations, and optimization settings. Public upstream revisions link to their exact commits.

The remaining DPH-generation settings appear in Table S5.

DPH generation terminates after 500–1,850 iterations under the early-stopping rule in Table S5. Gabor position and scale rates decay from 10^{-3} to 10^{-5}, while rotation and wave rates remain 10^{-3} and color remains 10^{-4}. The accompanying configurations specify the remaining optimizer and dependency details.

Table S3. Five-source coefficient comparison. Values are source means. Serialized PDA-PSNR is the primary endpoint. Runtime is elapsed fitting time per source.

Form	Steps	Continuous PDA-PSNR (dB)	Serialized PDA-PSNR (dB)	Runtime (s)
Factorized $\mathbf{c}, a$	2,500	44.462	43.270	102.1
Independent $\mathbf{b}$	2,500	42.815	42.238	184.2
Factorized $\mathbf{c}, a$	5,000	45.678	43.952	205.7
Independent $\mathbf{b}$	5,000	44.872	43.722	364.5
Factorized $\mathbf{c}, a$	10,000	46.019	44.414	406.8
Independent $\mathbf{b}$	10,000	45.687	44.347	717.3

Table S4. Implementations and optimization settings. Primitive counts are per checkerboard component.

Method	Pinned source	Main adaptation	Optimization
DPH generation	Experiment revision `e9ae1007c7f2`	RGB double-phase constraint, stored as 8-bit phase	AdamW, cosine schedule, at most 2,000 iterations with early stopping
GaussianImage	upstream `db9cead79a74`	Corrected packed-symmetric Cholesky backward, sinusoidal output, and 16k primitives/component	Adan, learning rate 10^{-3}, 10,000 steps
GS	**upstream `78ee9ce8e3ff`**	**16-entry frequency bank, top two active atoms, and 16k primitives/component**	**Adam, 10,000 steps, with accompanying parameter-group schedules**
INRSteg	upstream `4b95b1d6108d`	Per-case RGB adaptation, cover MLP 6×256, payload MLP 6×128, and no pretrained checkpoint	Adam, learning rate 10^{-4}, 10,000 steps
RoSteALS	upstream `8c09b022a362`	100-bit checkpoint (epoch 17, step 449,999), no fine-tuning, and a $512 \to 256 \to 512$ residual path	Inference only

Table S5. DPH generation configuration. Dimensions are reported as width × height.

Setting	Value
Input preparation	Aspect-ratio-preserving bilinear resize, centered zero padding to 4094×2400 pixels
Spatial light modulator (SLM)	Jasper Display Educational 4k with $3.74\,\mu$m pixel pitch
Color channels	639, 515, and 473 nm in three frames
Aperture	Centered circular aperture, diameter 1800 SLM pixels
Propagation	Forward band-limited angular spectrum
Depth planes	Three planes spanning 5 mm: -2.5, 0, and $+2.5$ mm. The flat-depth target is assigned to $+2.5$ mm
Target construction	Defocus-aware multiplane target with blur size 20 and blur ratio 3
Initialization	Zero phase initialization with deterministic per-source seeds recorded in the accompanying settings
Optimization	AdamW, learning rate 0.05 to 0.0005 with cosine annealing, mean reduction, and at most 2,000 iterations
Early stopping	Focused-plane reconstruction PSNR checked every 50 iterations, with a minimum of 500 iterations, patience 250 iterations, and minimum improvement 0.01 dB
Output	RGB double-phase constraint with phase quantized to 8 bits

Table S6. GaussianImage sensitivity comparison at the parameter-specific perturbations used for comparison. Values are median reductions in focused-plane reconstruction PSNR from the clean fitted model over the 30-source, 360-patch analysis. Lower is better.

Parameter	Perturbation	High-value drop [dB]	Low-value drop [dB]
Feature coefficient	0.05σ	0.032	0.275
Covariance log-area	$\times 1.05$	0.147	**0.117**
Anisotropy	$\times 1.25$	0.272	0.360
Rotation	$5°$	0.381	0.583
Position	1 pixel	5.300	4.573

All experiments ran on one NVIDIA GeForce RTX 5090. The DPH-generation stack used Python 3.14.4, PyTorch 2.12.0+cu130, and Odak [28]. Method fitting and baseline evaluation used isolated environments recorded in the accompanying configurations, including Python 3.10 and CUDA 12.8. GaussianImage, GS, and INRSteg are fitted and stored per case. RoSteALS instead uses one pretrained network shared across all cases. These states therefore describe different storage and computational roles. Reported GaussianImage and GS fitting times are sequential wall-clock measurements on this hardware. GaussianImage time sums its two independent 10,000-step component fits, whereas GS time covers the joint two-component fit. Under these settings, a 10,000-step GS fit takes about 350 s, whereas the two GaussianImage component fits take 32.9 s in total. Post-fitting evaluation includes scheduling, QIM, saving and reloading, rendering, holographic reconstruction, and metric evaluation. These heterogeneous totals characterize the reported pipelines rather than an architecture-level timing ranking. Our profiling attributes much of the fitting-time gap to the GS backend evaluating more primitive–pixel pairs, whereas GaussianImage uses tiled culling. Reducing this implementation gap remains future work.

S4. CARRIER SELECTION AND RECOVERY PROTOCOL

S4.1. Carrier sensitivity and selected carriers

The sensitivity analyses compare candidate carrier statistics by measuring phase and focused-plane change after controlled perturbations. Their absolute values are not compared across representations because the parameter definitions, perturbation magnitudes, masks, and source cohorts differ. Tables S6 and S7 report these representation-specific comparisons.

The GaussianImage sensitivity analysis uses 30 regenerated sources and 12 phase patches per source. For each patch and component, it deterministically perturbs a subset equal to 5% of the valid primitives, sampled from the eligibility pool shared by all candidate parameters, with three subset seeds. For the feature coefficient, σ is the robust within-patch/component scale based on the interquartile range (IQR), $\mathrm{IQR}/1.349$, with the standard deviation used only when the IQR is zero. The log-area perturbation is implemented as $\mu' = \mu \pm \log(1.05)$. It therefore multiplies both covariance eigenvalues by 1.05 or its reciprocal while preserving anisotropy and orientation. It does not multiply the stored log statistic μ.

The GS sensitivity analysis uses eight fixed sources and subset seeds 7, 42, and 2026. The accompanying reproducibility files record their identifiers and assignments. Each source modifies 100 eligible primitives without replacement. The phase results aggregate all eight sources and three seeds, while the focused-plane results use the same sources with seed 42.

Color and active-wave amplitudes are distortion-equivalent because they enter the rendered Gabor primitive multiplicatively. The selected carrier statistic is therefore the log Frobenius norm of the active coefficient block $\mathbf{B} = [\mathbf{b}_k]_{k\in\mathcal{K}}$. Under the stored factorization $\mathbf{b}_k = a_k\mathbf{c}$, this is exactly $\rho = \log(\|\mathbf{c}\|_2\|\mathbf{a}_\mathcal{K}\|_2)$. Table S8 lists the selected one-branch policies. The component differences are small. These choices are fixed engineering decisions rather than evidence of intrinsic component robustness.

After these parameter analyses, component selection and final policy confirmation used the 50-source DIV2K development cohort. The centered samples from this cohort contribute 50 sources to the expanded four-method benchmark and are therefore not independent validation of the carrier-policy choice. The additional 100 CLIC and 50 DIV2K sources were selected without method-result information and use the already fixed carrier policy.

Table S7. GS sensitivity at the largest tested perturbation for each parameter, relative to the clean fit. The columns report continuous PDA-PSNR and focused-plane reconstruction PSNR. Higher is better.

Parameter	Perturbation	PDA-PSNR [dB]	Focused PSNR [dB]
Wave amplitude	$\times 1.02$	86.971	63.061
Color amplitude	$\times 1.02$	86.971	63.061
Anisotropy	$\times 1.02$	79.267	59.849
Envelope log-area	$\times 1.02$	79.145	59.695
Rotation	$0.50°$	78.713	60.263
Position	0.50 pixel	71.599	57.590

Table S8. Selected carrier policies and eligibility rules. Component contrasts are source-level focused-plane PSNR differences with 95% confidence intervals (CIs). Positive values favor the selected component.

Method	Selected carrier	Component evidence	Eligibility rule
GaussianImage	Covariance log-area μ, low-value component	Low minus high: $+0.149$ dB, 95% CI $[+0.025, +0.270]$, Holm $p = 0.0625$	Finite physical position and Cholesky state
GS	**Effective log amplitude ρ, high-value component**	**High minus low: $+0.061$ dB, 95% CI $[-0.210, +0.329]$, Holm $p = 0.9813$**	**Finite state, effective amplitude at least 10^{-12}, and unchanged active top-two indices**

Table S9. Canonical primitive identity fields.

Method	Included in identity	Excluded from identity
GaussianImage	Component and physical x, y position	Cholesky state, log-area, coefficient, importance, array index
GS	**Component, x, y position, two log-scales, rotation, and sorted active-frequency indices**	**Color and wave amplitudes, ρ, array index**

S4.2. Keyed schedule and recovery

Schedule reconstruction. Primitive array indices are not stable identities because serialization or reordering can change them. Table S9 lists the embedding-invariant fields used to define each primitive identity.

Each identity is the Secure Hash Algorithm 256-bit (SHA-256) digest of a canonical byte record. Primitives that do not satisfy the eligibility rules and all members of a duplicated identity class are excluded. The schedule uses the Hash-based Message Authentication Code (HMAC) and the HMAC-based Extract-and-Expand Key Derivation Function with SHA-256 (HKDF-SHA256). HKDF-SHA256 expands the shared key and public schedule nonce into independent selection, ordering, and dither keys. HMAC-derived ranks select B eligible primitives for a B-bit payload and determine their order and centered dithers in $[-\Delta/2, \Delta/2)$. The decoder recomputes this schedule from the saved state and shared key, so it requires no stored carrier manifest.

Self-contained container. Clean and stego representations use the same container and public-header schema. Both carry a fixed-size recovery descriptor protected by the Advanced Encryption Standard with a 256-bit key in Galois/Counter Mode (AES-256-GCM): a 16-byte plaintext and 16-byte authentication tag form a 32-byte ciphertext, without exposing payload presence or length in plaintext. A random 12-byte descriptor nonce is distinct from the 16-byte schedule nonce. Authenticated associated data bind the descriptor to the canonical public fields and to the length and SHA-256 digest of the serialized primitive state. Given the container and 256-bit shared key, the decoder authenticates the descriptor and reconstructs the schedule without external per-case metadata.

Validation and scope. The standard decoder recovered all 4,500 embedded payloads after save and reload, rejected tested wrong keys, and returned `no_payload` for clean containers. The CLIC evaluations left every serialized phase sample outside the centered patch unchanged. A separate 2,100-state validation confirmed byte-exact primitive-state preservation and authenticated wrong-key rejection. Container overhead was 1,051–1,131 bytes. These checks establish saved-state recovery and container integrity.

S4.3. Representation robustness and keyless detection

Reordering and reduced precision. We jointly permute all primitive arrays and recover every 100- and 1,000-bit payload across 200 sources and three keys. Unrelated keys produce bit-error rates of 0.498 and 0.500, respectively, and every authenticated descriptor rejects the wrong key. For reduced-precision storage, we quantize the position, scale, and rotation fields before computing primitive identities and save all floating-point fields at 16-bit precision. The decoder again recovers every payload. On 20 sources at 100 bits, this conversion lowers end-to-end reconstruction PSNR by 0.50 dB (95% confidence interval [0.36, 0.65] dB) relative to the full-precision representation. The clean and stego reconstructions after conversion differ by 74.81 dB, so the additional error comes primarily from reduced precision rather than embedding.

Keyless detection. We train source-disjoint classifiers on 200 sources, keeping every clean and stego variant of a source in the same fold. The parameter-domain tests use predefined statistics or a multilayer perceptron applied to summaries of the saved primitive parameters. We also apply the published SRNet image steganalysis architecture [29] to the stored phase. SRNet evaluates each phase channel and each nonoverlapping 256×256 quadrant, and its predictions are averaged for the 512×512 patch. At 100 and 1,000 bits, the evaluated classifiers do not distinguish clean and stego outputs (Table S10). Dense coefficient-scaling controls confirm sensitivity to larger changes: they reach AUC 0.746 in the parameter domain and 0.897 in the stored phase. Actual QIM changes 0.028% and 0.097% of stored phase samples at 100 and 1,000 bits, compared with 49.0% for the phase control. These results apply only to the evaluated classifiers and do not establish universal resistance to steganalysis.

Table S10. Source-disjoint keyless detection and sensitivity controls.

Domain	Condition	ROC-AUC	95% CI
Parameters	Statistics, 100 bits	0.501	[0.496, 0.505]
Parameters	Statistics, 1,000 bits	0.509	[0.495, 0.523]
Parameters	Summary classifier, 100 bits	0.500	[0.500, 0.501]
Parameters	Summary classifier, 1,000 bits	0.500	[0.500, 0.501]
Parameters	5% coefficient scaling	0.746	[0.719, 0.776]
Phase	SRNet, 100 bits	0.500	[0.499, 0.501]
Phase	SRNet, 1,000 bits	0.500	[0.500, 0.501]
Phase	5% coefficient scaling	0.897	[0.866, 0.925]

Primitive pruning. The original decoder ranks all eligible primitives globally. Deleting even an unselected primitive can therefore shift later bit assignments. For the pruning experiment, a keyed hash of each primitive identity assigns one Reed–Solomon-coded bit to a group of primitives, so deletion affects only that group. We encode the 1,000-bit payload with a Reed–Solomon code, producing 2,040 coded bits. We prune all per-primitive arrays either randomly or in increasing order of effective amplitude. Across 200 sources and three keys, the decoder recovers every payload through 1% pruning. At 5%, it recovers all 600 randomly pruned cases and 599 of 600 amplitude-pruned cases (Table S11).

Table S11. Exact 1,000-bit recovery with identity-based assignment and Reed–Solomon coding. Each cell reports random/lowest-amplitude pruning over 600 source–key conditions.

Pruned primitives	0.1%	0.5%	1%	5%
Exact payloads	600/600	600/600	600/600	600/599
Recovery rate	100/100%	100/100%	100/100%	100/99.8%

The protected condition modifies 2,040 of 16,000 high-value primitives. Its mean PDA-PSNR, reconstruction PSNR, SSIM, and LPIPS-Alex are 45.64 dB, 43.01 dB, 0.974, and 0.029, matching the main 100-bit GS result at the reported precision. This Reed–Solomon protocol is a robustness extension and is not used in the main benchmark.

S5. QIM CALIBRATION AND PAYLOAD STATISTICS

Three predefined QIM keys are used after fitting. Their exact values are included in the accompanying reproducibility files. The same fitted representation is reused for all three keys, and payload-study metrics aggregate the three keys within each source before source-level comparison.

Using 10 calibration sources, we selected a shared multiplicative QIM factor γ, which sets the log-domain lattice spacing to $\Delta = \log \gamma$. We tested $\gamma = 1.0025$, 1.005, 1.01, and 1.02. A candidate had to retain the requested payload, decode without error after the model was saved and reloaded, and change no serialized phase samples outside the selected patch. Among eligible candidates, 1.0025 achieved the highest source-aggregated reconstruction PSNR between the stego output and its clean fitted baseline at 1,000 bits. The remaining 15 sources were reserved for the paired payload-level comparison. Embedding-only fidelity decreases with payload, whereas end-to-end fidelity changes little (Fig. S1).

On the reserved validation set, the transition from GaussianImage to GS occurs between 100 and 250 bits for focused-plane reconstruction PSNR measured relative to each method's clean fitted baseline (Table S12).

We then apply the fixed DIV2K protocol, including its carrier policy, QIM factor, checkpoint, payload, and keys, to 100 CLIC2020 Professional validation sources using one centered patch per source, without reselection. The 250-bit crossover is reproduced in this cross-dataset validation (Table S13).

The CLIC validation also supports the representation-dominance interpretation. Across payloads, embedding-only focused-plane PSNR exceeds end-to-end PSNR by 26.40–36.17 dB for GS and 27.56–43.02 dB for GaussianImage. Increasing the payload from 25 to 1,000 bits changes mean end-to-end focused-plane PSNR by less than 0.001 dB for either method.

S6. TRAINING OBJECTIVES AND ABLATIONS

The 20-source objective ablation selects the Circular-trained model used in the 200-source benchmark.

At 10,000 steps, Circular increases continuous-phase PDA-PSNR over MSE by +1.088 dB (95% paired Student-t interval [+0.800,+1.375] dB) and is higher on 19 of 20 sources. After serialization, Circular improves mean PDA-PSNR from 44.892 to 45.541 dB and mean PDA-SSIM from 0.98716 to 0.98878. Its paired stored-phase gains are +0.649 dB in PDA-PSNR (95% confidence interval [+0.481,+0.817] dB, 20 of 20 sources) and +0.00161 in PDA-SSIM (95% confidence interval [+0.00106,+0.00217], 18 of 20 sources). Circular ranks above MSE for both the continuous- and stored-phase PDA endpoints.

The propagation-aware reference is

$$\mathcal{L}_{\mathrm{circ+focus}} = \frac{1}{2}\left(\mathcal{L}^{\mathrm{hi}}_{\mathrm{circ}} + \mathcal{L}^{\mathrm{lo}}_{\mathrm{circ}}\right) + 0.05\,\mathcal{L}_{\mathrm{focus}}. \tag{S6}$$

For component s, $\mathcal{L}^{s}_{\mathrm{circ}}$ averages the periodic phase error over every spatial sample $\mathbf{x} \in \Omega$ and all K phase channels. The one-half factor therefore gives the same equal-component average as the global circular objective in the main paper. Both component models are optimized jointly. The implementation's continuous cycle-normalized predictions, which are equivalent to the physical phases reported in the main paper, are recomposed into a 512×512 RGB phase patch and inserted

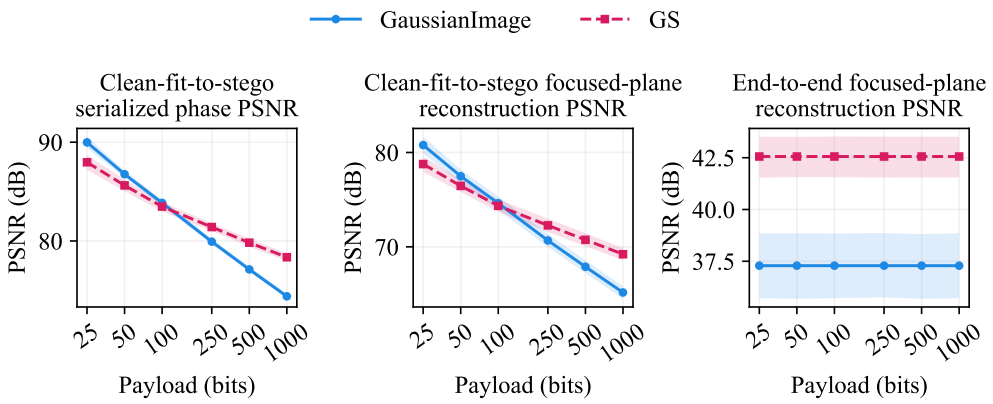


Fig. S1. Payload-size fidelity. Results cover the 25-source DIV2K cohort. Left: clean-fit-to-stego serialized phase PSNR. Middle: clean-fit-to-stego focused-plane reconstruction PSNR. Right: end-to-end focused-plane reconstruction PSNR relative to the clean DPH. Shading shows descriptive 95% source-bootstrap intervals. Paired tests use the reserved 15-source subset.

Table S12. Source-paired payload contrasts on the reserved 15-source validation set. Positive differences favor GS. Confidence intervals are pointwise, unadjusted 95% source-bootstrap intervals. The p-values use paired two-sided sign-flip tests with Holm adjustment across the six payloads.

Bits	Gabor − GaussianImage [dB]	95% CI [dB]	Raw p	Holm p
25	-1.902	$[-3.158, -0.692]$	0.01050	0.04200
50	-1.200	$[-2.352, -0.105]$	0.06275	0.12549
100	-0.457	$[-1.388, +0.449]$	0.36178	0.36178
250	$+1.448$	$[+0.452, +2.369]$	0.01565	0.04695
500	$+2.803$	$[+1.845, +3.671]$	0.00050	0.00250
1000	$+3.973$	$[+3.103, +4.799]$	0.00010	0.00060

Table S13. Cross-dataset validation contrasts over 100 centered CLIC sources. Positive differences favor GS. Confidence intervals and tests use the same source-level procedures as Table S12.

Bits	Gabor − GaussianImage [dB]	95% CI [dB]	Raw p	Holm p
25	-1.752	$[-2.271, -1.237]$	< 0.0001	0.00030
50	-0.921	$[-1.380, -0.489]$	< 0.0001	0.00030
100	$+0.164$	$[-0.307, +0.601]$	0.48543	0.48543
250	$+1.543$	$[+1.093, +1.973]$	< 0.0001	0.00030
500	$+2.801$	$[+2.331, +3.240]$	< 0.0001	0.00030
1000	$+3.945$	$[+3.480, +4.370]$	< 0.0001	0.00030

at the target location in the otherwise clean full hologram. No modulo-one canonicalization or 8-bit quantization is applied inside the optimization loop.

After propagation to $+2.5$ mm, let R denote the 512×512 focused-plane ROI. Its RGB intensity MSE is

$$\mathcal{L}_{\text{focus}} = \frac{1}{3|R|} \sum_{c=1}^{3} \sum_{\mathbf{x} \in R} \left[I_{\text{pred},c}(\mathbf{x}) - I_{\text{clean},c}(\mathbf{x}) \right]^2 . \tag{S7}$$

The clean and predicted intensities are clamped to $[0, 1]$ but are not normalized independently. The clean path is detached, whereas gradients pass through patch insertion, propagation of the full hologram, and the candidate intensity. Because serialization is outside this path, no straight-through estimator or quantization-gradient approximation is used.

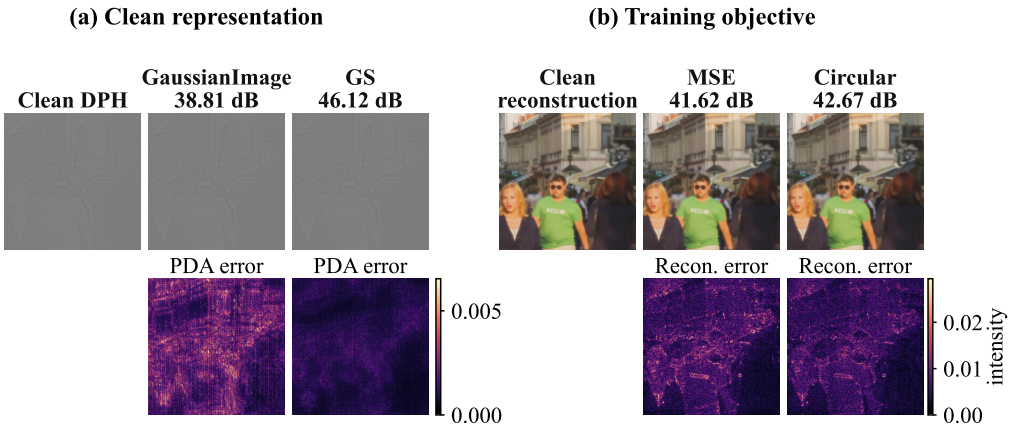


Fig. S2. Representative comparison on DIV2K 0173. (a) Pre-serialization phase fits and PDA-aligned phase-error maps for GaussianImage and GS. Annotations report PDA-PSNR; the error maps use a shared square-root color scale. (b) Focused-plane reconstructions and RGB error maps for GS trained with MSE or Circular. Annotations report focused-plane PSNR.

Table S14. Dataset-stratified focused-plane reconstruction PSNR at 100 bits. The last column reports GS minus RoSteALS.

Cohort	N	INRSteg	GaussianImage	**GS**	RoSteALS	GS–RoSteALS [95% CI]
CLIC validation	100	20.05	38.07	**43.17**	41.19	$+1.978$ $[+1.614, +2.326]$
DIV2K validation	50	20.14	37.48	**42.90**	40.65	$+2.259$ $[+1.685, +2.828]$
DIV2K development	50	20.45	37.92	**42.80**	40.08	$+2.724$ $[+2.259, +3.154]$

S7. CENTERED CROSS-DATASET FOUR-METHOD VALIDATION

The expanded 100-bit benchmark uses one centered 512×512 patch from each of 200 distinct natural-image sources: 100 CLIC sources, 50 newly selected DIV2K sources, and the 50-source DIV2K development cohort. The DIV2K validation and development cohorts do not overlap. The validation cohorts were evaluated after all method choices were fixed, whereas the DIV2K development cohort contributed to method and carrier-policy selection. The source cohorts were fixed before method fitting, and no source was replaced based on fidelity, bit-error rate, or method failure.

The decision to pool these three natural-image cohorts was made after inspecting dataset-level results. The pooled analysis is therefore reported as a transparently disclosed post-result scope decision rather than a preregistered analysis.

The dataset-stratified primary optical endpoint appears in Table S14. Explicit-method keys are aggregated in MSE space within each source before conversion to PSNR. The interval is a 20,000-draw paired source-bootstrap interval with seed 20260823.

Across the 150 CLIC and DIV2K validation sources, GS remains $+2.071$ dB above RoSteALS (95% source-bootstrap CI $[+1.764, +2.377]$ dB) and is higher on 127 sources. Across all 200 sources, GS exceeds RoSteALS by $+2.235$ dB (95% source-bootstrap CI $[+1.975, +2.489]$ dB, 173/200 wins), while clean GS exceeds GaussianImage-16k by $+7.645$ dB in pre-serialization PDA-PSNR and $+5.124$ dB in focused-plane reconstruction PSNR.

GaussianImage, GS, and INRSteg recover every embedded payload using their declared recovery inputs. RoSteALS makes one bit error in one CLIC source, corresponding to a pooled bit-error rate of 5×10^{-5} over 20,000 payload bits. These bit-error rates do not imply equivalent attack robustness because the decoder inputs differ as described in the main paper.

S7.1. Equal-parameter comparisons

The main benchmark matches 16,000 primitives per checkerboard component. For equal-parameter comparisons, we count parameters in the final stored representation. We evaluate GaussianImage-24k versus GS-16k at 384,000 parameters each, and GaussianImage-16k versus GS-10.667k at 256,000 versus 256,008 parameters. The latter differs by only 0.0031%. The count includes two

frequency-selection indices per GS primitive but does not match serialized bytes, runtime, or computation. All other experimental settings remain fixed.

Before embedding, clean GS-16k exceeds clean GaussianImage-24k by 8.060 dB in PDA-PSNR (95% source-bootstrap CI [7.798, 8.331] dB, 200/200 wins) and 5.441 dB in holographic-reconstruction PSNR (95% CI [5.221, 5.662] dB, 199/200 wins).

Table S15. Primitive-count and equal-parameter controls at 100 bits over 200 sources.

Method	Primitives/component	Parameters/patch	PDA-PSNR	Recon. PSNR	SSIM	LPIPS
GaussianImage-16k	16,000	256,000	40.17	37.89	0.917	0.138
GaussianImage-24k	24,000	384,000	39.78	37.57	0.910	0.143
GS-16k	16,000	384,000	**45.64**	**43.01**	**0.974**	**0.029**
GS-10.667k	10,667	256,008	**44.70**	**42.00**	**0.966**	**0.044**

At 384,000 parameters, GS-16k exceeds GaussianImage-24k by 5.856 dB in PDA-PSNR (95% source-bootstrap CI [5.703, 6.014] dB, 200/200 wins) and 5.441 dB in holographic-reconstruction PSNR (95% CI [5.224, 5.663] dB, 199/200 wins). In the reverse match, GS-10.667k exceeds GaussianImage-16k by 4.525 dB in PDA-PSNR (95% CI [4.337, 4.711] dB, 199/200 wins) and 4.117 dB in reconstruction PSNR (95% CI [3.870, 4.360] dB). It also improves SSIM by 0.049 (95% CI [0.044, 0.054]) and reduces LPIPS-Alex by 0.095 (95% CI [0.086, 0.104]). Both pairs recover all payloads without bit errors.

S7.2. Phase-distribution-aligned four-method comparison

For the primitive-matched 200-source, 100-bit benchmark, GS has the highest mean PDA-PSNR and PDA-SSIM (Table S16). Section S2.2 defines the metrics, and Sec. S2 defines their aggregation.

Table S16. Source-paired PDA differences at 100 bits over 200 sources. Differences are GS minus the named baseline. Intervals are pointwise 95% source-bootstrap intervals. Wins count sources with a positive difference.

Baseline	Metric	Mean difference	95% CI	Wins
INRSteg	PDA-PSNR (dB)	+21.194	[+20.862,+21.517]	200/200
GaussianImage-16k	PDA-PSNR (dB)	+5.468	[+5.266,+5.665]	199/200
RoSteALS	PDA-PSNR (dB)	+4.842	[+4.704,+4.982]	200/200
INRSteg	PDA-SSIM	+0.25047	[+0.24732,+0.25351]	200/200
GaussianImage-16k	PDA-SSIM	+0.02703	[+0.02540,+0.02867]	199/200
RoSteALS	PDA-SSIM	+0.01351	[+0.01284,+0.01420]	200/200

REFERENCES

1. C. K. Hsueh and A. A. Sawchuk, "Computer-generated double-phase holograms," Appl. Opt. **17**, 3874–3883 (1978).
2. X. Fan, Y. Zhan, A. Mazumdar, and K. Akşit, "Compressing double-phase holograms using 2D gaussians," in *Eurographics 2026 - Posters*, T. Gerrits and M. Teschner, eds. (The Eurographics Association, 2026).
3. K. Kavaklı, L. Shi, H. Ürey, *et al.*, "Multi-color Holograms Improve Brightness in Holographic Displays," in *SIGGRAPH Asia 2023 Conference Papers*, (Association for Computing Machinery, New York, NY, USA, 2023), SA '23.
4. Z. Peng, Y. Zhan, J. Spjut, and K. Akşit, "Assessing learned models for phase-only hologram compression," in *SIGGRAPH Posters '25: Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters*, (ACM, New York, NY, USA, 2025).
5. X. Zhang, X. Ge, T. Xu, *et al.*, "GaussianImage: 1000 FPS Image Representation and Compression by 2D Gaussian Splatting," in *Computer Vision – ECCV 2024*, vol. 15067 of *Lecture Notes in Computer Science* (Springer, Cham, 2024), pp. 327–345.
6. S. W. Wurster, R. Zhang, and C. Zheng, "Gabor Splatting for High-Quality Gigapixel Image Representations," in *ACM SIGGRAPH 2024 Posters*, (Association for Computing Machinery, New York, NY, USA, 2024), SIGGRAPH '24.
7. H. Hamam, "Digital holography-based steganography," Opt. Lett. **35**, 4175–4177 (2010).

8. W. Zou, Z. Zhuang, S. Jiao, *et al.*, "Image steganography based on digital holography and saliency map," Opt. Eng. **58**, 013102 (2019).
9. H. Kim and Y. H. Lee, "Optimal watermarking of digital hologram of 3-d object," Opt. Express **13**, 2881–2886 (2005).
10. W.-J. Hwang, H.-T. Chan, and C.-J. Cheng, "Hologram authentication based on a secure watermarking algorithm using cellular automata," Appl. Opt. **53**, G64–G73 (2014).
11. J.-W. Kang, J.-E. Lee, J.-H. Choi, *et al.*, "Digital hologram watermarking based on multiple deep neural networks training reconstruction and attack," Sensors **21**, 4977 (2021).
12. T. Shimobaba, Y. Endo, R. Hirayama, *et al.*, "Holographic microinformation hiding," Appl. Opt. **56**, 833–837 (2017).
13. P. W. M. Tsang and T.-C. Poon, "Data-embedded-error-diffusion hologram," Chin. Opt. Lett. **12**, 060017 (2014).
14. X. Wang, Q. Zhou, L. Zhang, *et al.*, "Computational imaging encryption with a steganographic and holographic authentication strategy," Laser & Photonics Rev. **18**, 2300820 (2024).
15. X. Jiang, X. Li, S. Zhang, *et al.*, "WO_x/Li^+–Ag^+/TiO_2 Optoelectronic Memory for Holographic Steganography," Opt. Lett. **51**, 2720–2723 (2026).
16. B. Chen and G. W. Wornell, "Quantization index modulation: A class of provably good methods for digital watermarking and information embedding," IEEE Trans. on Inf. Theory **47**, 1423–1443 (2001).
17. Y. Uchida, Y. Nagai, S. Sakazawa, and S. Satoh, "Embedding watermarks into deep neural networks," in *Proceedings of the 2017 ACM on International Conference on Multimedia Retrieval*, (Association for Computing Machinery, 2017), pp. 269–277.
18. T. Wu, R. Cheng, X. Fan, and K. Akşit, "Gaussian image steganography via parameter-domain keyed embeddings," in *SIGGRAPH Asia 2026 Technical Communications*, (Association for Computing Machinery, New York, NY, USA, 2026).
19. L. Qiu Lin, H. Kam, Q. Sun, and K. Akşit, "Foveation improves payload capacity in steganography," in *Proceedings of the SIGGRAPH Asia 2025 Posters*, (Association for Computing Machinery, New York, NY, USA, 2025), SA Posters '25, pp. 1–2.
20. X. Huang, R. Li, Y.-m. Cheung, *et al.*, "GaussianMarker: Uncertainty-aware copyright protection of 3D gaussian splatting," in *Advances in Neural Information Processing Systems*, vol. 37 (Curran Associates, Inc., 2024).
21. Y. Jang, H. Park, F. Yang, *et al.*, "3D-GSW: 3d gaussian splatting for robust watermarking," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, (IEEE, Los Alamitos, CA, USA, 2025), pp. 5938–5948.
22. Z. Chen, G. Wang, J. Zhu, *et al.*, "GuardSplat: Efficient and robust watermarking for 3d gaussian splatting," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, (IEEE, Los Alamitos, CA, USA, 2025), pp. 16325–16335.
23. S. Song, S. Yang, C. D. Yoo, and J. Kim, "Implicit Steganography Beyond the Constraints of Modality," in *Computer Vision – ECCV 2024*, vol. 15144 of *Lecture Notes in Computer Science* A. Leonardis, E. Ricci, S. Roth, *et al.*, eds. (Springer, Cham, 2025), pp. 289–304.
24. T. Bui, S. Agarwal, N. Yu, and J. Collomosse, "RoSteALS: Robust Steganography using Autoencoder Latent Space," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, (IEEE, 2023), pp. 933–942.
25. S. Choi, T. H. Lee, J. Y. Cha, *et al.*, "Phase distribution matters: On the importance of phase distribution alignment (PDA) in holographic applications," in *Proceedings of the 33rd ACM International Conference on Multimedia*, (Association for Computing Machinery, 2025), pp. 8930–8938.
26. Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," IEEE Trans. on Image Process. **13**, 600–612 (2004).
27. R. Zhang, P. Isola, A. A. Efros, *et al.*, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, (2018), pp. 586–595.
28. K. Akşit, J. Beyazian, P. Chakravarthula, *et al.*, "Odak," Accessed 24 August 2026.
29. M. Boroumand, M. Chen, and J. Fridrich, "Deep residual network for steganalysis of digital images," IEEE Trans. on Inf. Forensics Secur. **14**, 1181–1193 (2019).